TUXIVISION

TS5M02 Camera Module Specification

Product Information

Model: TS5M02
Type: 5MP Auto Focus USB Camera Module
Interface: USB2.0
Output: MJPG / YUYV
Resolution: 2592×1944
Focus: Auto Focus

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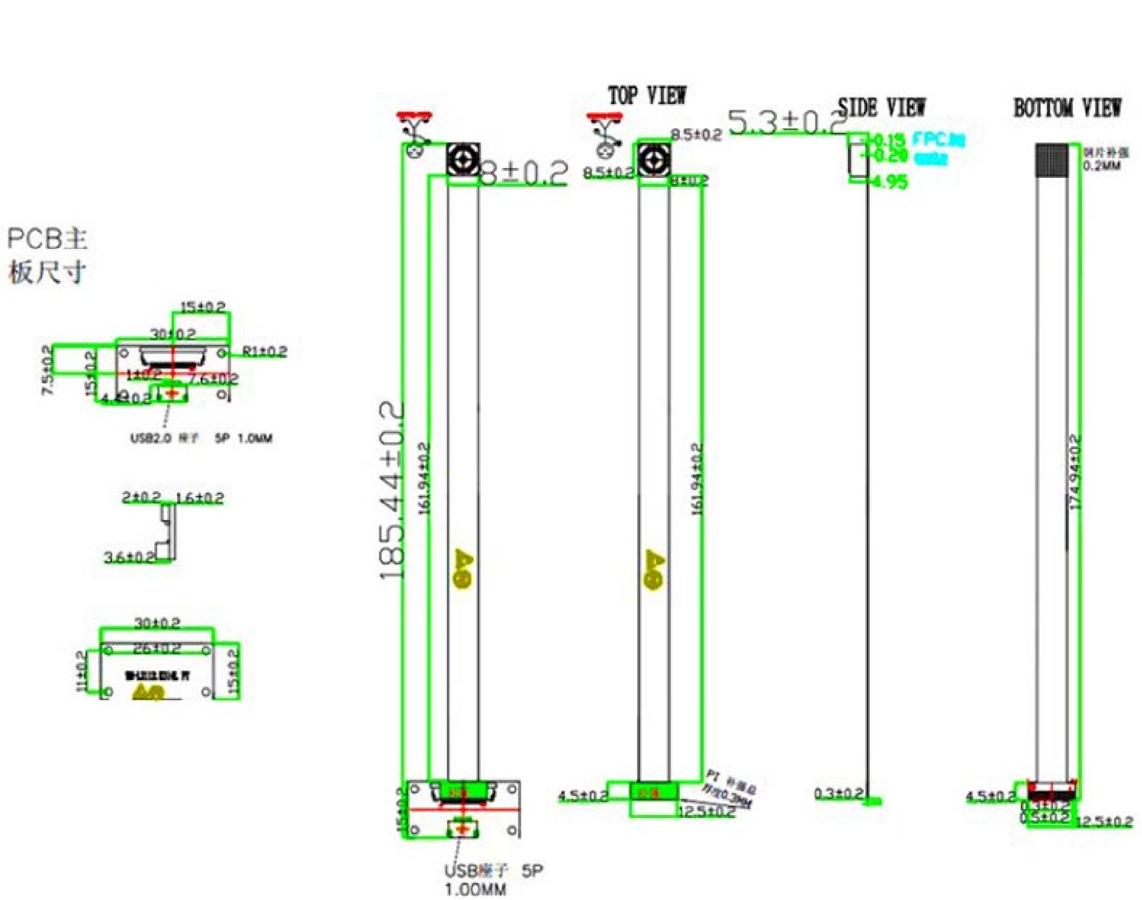
Main Specifications

Module No	**TS5M02**
Sensor	OV5693 CMOS
Sensor Size	1/4"
Pixel Size	1.4µm × 1.4µm
Focus Type	Auto Focus (AF)
Effective Resolution	2592 × 1944 (5MP)
Output Format	MJPG / YUYV
Interface	USB 2.0
Lens	60° Distortion-Free Lens
Auto Exposure	Supported
Auto White Balance	Supported
Auto Gain Control	Supported

Adjustable Parameters	Brightness, Contrast, Hue, Saturation, Sharpness, Gamma, White Balance, Backlight Compensation, Gain, Exposure, Low-Light Compensation
Power Supply	USB 2.0 4P
Operating Voltage	DC 5V
Operating Current	≤160mA
Operating Temperature	-10°C ~ +70°C
Storage Temperature	-20°C ~ +85°C
Supported Systems	Windows XP/7/8/10/11, Linux (UVC), macOS, Android UVC
Resolution & Frame Rate	2592×1944@30FPS
Module Dimensions	61 × 9 mm

Mechanical Drawing



Appearance Specification

No.	Inspection Item	Requirement	Inspection Method	Defect Classification
1	**Module Appearance**	Module surface shall be free from contamination and scratches visible to the naked eye. No foreign particles are allowed inside plastic components.	8× Magnifier	A
2	**Lens**	Under illumination of **800 ± 100 Lux**, inspect by direct viewing and at a **45° reflected light angle**. No scratches, stains, or other defects visible to the naked eye are allowed.	8× Magnifier	A

3	**Thread-locking Glue**	Adhesive coverage shall be **1/3 to 1/2 of the lens circumference**. Glue shall not overflow onto the lens surface or lens holder/base.	8× Magnifier	A
4	**Sealant**	Sealant shall uniformly fill the gap between the base and the FPC. No gaps, uneven thickness, or overflow beyond the FPC edge that could cause dimensional nonconformance are allowed.	8× Magnifier	A
5	**PCB**	1. Surface shall be clean and free from contamination. Markings and printed characters shall be clear and legible. 2. Edge burrs: Length < 1.0 mm, Width < 0.3 mm. 3. Edge notches: Length < 1.0 mm, Width < 0.1 mm. Maximum 2 defects per edge.	45× Microscope (Continuous Zoom)	A

No.	Inspection Item	Requirement	Inspection Method	Defect Classification
1	Height	Must comply with drawing specifications.	Digital Caliper	A
2	Length	Must comply with drawing specifications.	Digital Caliper	A
3	Width	Must comply with drawing specifications.	Digital Caliper	A

Image Specification

No.	Inspection Item	Requirement	Inspection Method	Defect Classification
1	**Dust / Contamination**	Under white-field conditions, capture an image. No visible foreign particles or contamination are allowed across the entire image area.	Test fixture, light box, 6500K white backlight	A
2	**Imaging Direction**	Must comply with drawing/specification requirements.	Test fixture	A

3	**Dead Pixels & Cluster Defects**	**Dark-field:** Total bright pixels < 400 pixels; clusters of 2 adjacent bright pixels < 20; no clusters containing 3 or more bright pixels. **White-field:** Total dark pixels < 400 pixels; clusters of 2 adjacent dark pixels < 20; no clusters containing 3 or more dark pixels.	Test software, light box (6500K white backlight)	A
4	**Resolution**	5MP module: Center resolution ≥ 600 LW/PH.	MTF test software	A
5	**Field of View (FOV)**	Horizontal: According to drawing requirements. Vertical: According to drawing requirements. Diagonal: According to drawing requirements.	Test fixture, FOV target chart	A
6	**Shading / Image Uniformity**	Average brightness at the four corners (90% field position) must be ≥ 45% of the center brightness. Each corner brightness must be ≥ 35% of the center brightness. (Sampling window: 40×40 pixels; sensor vignetting compensation disabled.)	Test software, light box (3400 Lux, 6500K white backlight)	A
7	**Color Accuracy**	Color reproduction error ≤ 20%.	Test software, Color Checker Card	A
8	**TV Distortion (Geometric Distortion)**	Pincushion distortion (positive) or barrel distortion (negative) ≤ 1%.	Test software, ISO12233 chart	A
9	**Grayscale Performance**	Brightness difference between two adjacent distinguishable grayscale levels must be less than 8.	Test software, grayscale chart	A

10	**Chromatic Aberration (CA)**	CA value < 0.5.	Test software, ISO12233 chart	A
11	**Thread Torque (Torsional Strength)**	≥ 1 kgf·cm	Torque wrench	A
12	**Base Thrust Strength**	≥ 2 N	Force gauge	A

Explanation of Defect Classification

Class A – Critical Defect

A defect classified as Class A includes:

- *Batch rejection determined during inspection.*
- *Defects causing significant economic loss.*
- *Defects directly affecting product quality.*
- *Defects impacting major functions, performance, or technical specifications.*

Classification Code: A

Class B – Minor Defect

A defect is classified as Class B only if all of the following conditions are met:

1. *Does not affect the appearance quality of the final product.*
2. *Does not affect the technical performance of the final product.*
3. *Does not affect subsequent processing or assembly of raw materials or finished products.*

Classification Code: B

Sampling Plan

No.	Inspection Item	Sampling Frequency	Inspection Method	Remarks
Imaging & Reliability Items				
1	Contamination / Spots	N=5 per lot, C=0	Same as production standard	100% production inspection item
2	Image Orientation	N=5 per lot, C=0	Test fixture	100% production inspection item
3	Dead Pixels & Cluster Defects	N=5 per lot, C=0	Same as production standard	100% production inspection item
4	Resolution	N=5 per lot, C=0	MTF test	100% production

			software	inspection item
5	Field of View (FOV)	N=5 per lot, C=0	Test fixture, FOV chart	Evaluation item during sample stage; QA sampling inspection item
6	Image Uniformity	N=5 per lot, C=0	Test software, light box (3400 Lux, 6500K white backlight)	Evaluation item during sample stage; QA sampling inspection item
7	Chromatic Aberration	N=5 per lot, C=0	Test software, Color Checker Card	Evaluation item during sample stage; QA sampling inspection item
8	Geometric Distortion	N=5 per lot, C=0	Test software, ISO12233 chart	Evaluation item during sample stage; QA sampling inspection item
9	Gray Scale	N=5 per lot, C=0	Test software, grayscale chart	Evaluation item during sample stage; QA sampling inspection item
10	Reliability Test	Sample Development Stage: 2 pcs/batch, Acceptance Level 0/1; Stable Production Stage: 5 pcs/item/month, Acceptance Level 0/1	Reliability test procedures	QA sampling inspection item
Appearance Items				
1	Module Appearance	AQL = 0.65	8× Magnifier	100% production inspection item
2	Lens	AQL = 0.65	8× Magnifier	100% production

				inspection item
3	Thread-locking Glue	AQL = 0.65	8× Magnifier	100% production inspection item
4	Sealant	AQL = 0.65	8× Magnifier	100% production inspection item
5	PCB	AQL = 0.65	45× Microscope (Continuous Zoom)	100% production inspection item
6	Connector	AQL = 0.65	45× Microscope (Continuous Zoom)	100% production inspection item
Mechanical Dimensions				
1	Height	N=5 per lot, C=0	Same as production standard	100% production inspection item
2	Length	N=5 per lot, C=0	Same as production standard	100% production inspection item
3	Width	N=5 per lot, C=0	Same as production standard	100% production inspection item
4	Thread Torque	N=5 per lot, C=0	Torque Wrench	QA sampling inspection item
5	Base Push Force	N=5 per lot, C=0	Force Gauge	QA sampling inspection item

Reliability Test

No.	Test Item	Conditions & Requirements	Test Equipment
1	**Low Temperature Storage**	-30°C, 96 hours. Temperature ramp-up/down time: 30 min each	Low temperature chamber

2	**Low Temperature Operation**	-30°C, 96 hours. Temperature ramp-up/down time: 30 min each	Low temperature chamber
3	**High Temperature Storage**	80°C, 96 hours. Temperature ramp-up/down time: 30 min each	High temperature chamber
4	**High Temperature Operation**	70°C, 96 hours. Temperature ramp-up/down time: 30 min each	High temperature chamber
5	**High Temperature & High Humidity Storage**	60°C / 90% RH, 120 hours. Temperature ramp-up/down time: 30 min each	Temperature & humidity chamber
6	**High Temperature & High Humidity Operation**	40°C / 90% RH, 48 hours. Temperature ramp-up/down time: 30 min each	Temperature & humidity chamber
7	**Thermal Shock Test (Cold–Hot Cycle)**	-30°C ↔ 80°C (each dwell time: 30 min). Transition time between hot and cold < 5 min. Total: 10 cycles	Low/high temperature chambers
8	**Free Drop Test (after packaging)**	Drop height: 150 cm, 10 times	Wooden floor (oak surface)
9	**Vibration Test (after packaging)**	Frequency: 10 Hz → 200 Hz → 10 Hz Amplitude: 2 mm Direction: XYZ axes Duration: 2 hours	Vibration testing machine
10	**ESD Test (Power Off)**	150 pF / 330 Ω, ±8 kV air discharge. Measure 10 times	Electrostatic discharge tester